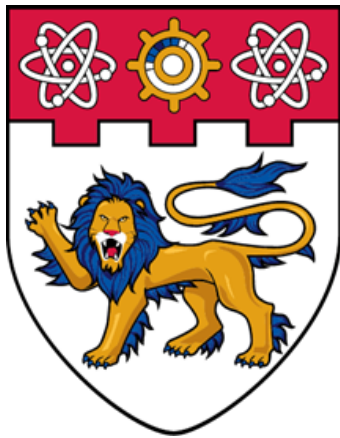




**NANYANG
TECHNOLOGICAL
UNIVERSITY**

SINGAPORE

SC2006 Software Engineering
Lab 1 Deliverables
Tutorial Group : SCE2

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Functional Requirements

1. User Input and Route Planning

- 1.1. The user must be able to input their starting and ending locations within Singapore.
 - 1.1.1. The system shall provide a text-based search bar for location input.
 - 1.1.2. The system must list a dropdown of auto-suggested locations based on user input in the search bar.
 - 1.1.3. The system must validate entered locations to ensure they are valid and within Singapore.
 - 1.1.4. The system must display an error message regarding invalid input and its reason within the dropdown.
 - 1.1.5. The system must have a clickable “search” button that invokes the generation of routes.
- 1.2. The system shall provide users with the option to select their mode (selectable tabs) of transport and their corresponding generated routes upon pressing the search button.
 - Driving
 - Public Transit (Bus and Train)
 - Walking
- 1.3. The system must generate and display possible routes from the starting point to the destination based on the selected mode of transport.
 - 1.3.1. The system must display individual routes in a list with information in accordance to the individual transport mode specs.
 - 1.3.2. The user must be able click onto any transport tab at any point to change the view list of generated routes.
 - 1.3.3. The system must call an external API (OneMap API) for route generation.
 - 1.3.4. The system must display a message of “No Routes Found” at the route's result page (each individual transport tab) upon failure to generate at least 1 route.
- 1.4. The system must calculate and display a convenience level score for each route based on the selected mode of transport.
 - 1.4.1. The score must be on a scale (1 - 10).
- 1.5. The user, if authenticated, must be able to save their planned routes to their profile.
- 1.6. The user's saved routes must be stored under the user's profile.

2. Driving Mode

- 2.1. The system must retrieve and display nearby carpark at the user specified destination.
 - 2.1.1. Carpark data must include the following information:
 - Parking space availability
 - Distance from destination
 - Parking Fee
- 2.2. The system shall display route results as a “route card” under the driving transport tab with information regarding reaching the destination by driving.
 - 2.2.1. The system must provide an estimated time of arrival (ETA) based on real-time traffic data.
 - 2.2.2. The system shall use real-time traffic conditions from external APIs.
 - 2.2.3. The system must modify the ETA based on traffic conditions.
 - 2.2.4. The system must display the traffic conditions on each “route card”.
- 2.3. The system must account for traffic incidents.
 - 2.3.1. The system must display traffic incidents as a pin and inline with the “route card”:
 - (Incident) along (Road name)
 - No Incidents Reported
 - 2.3.2. The system must be able to modify ETA based on traffic incidents.

3. Public Transit Mode

Results Display

- 3.1. The system must display route results as a “route card” under the public transport tab with information regarding reaching the destination by public transport.
- 3.2. The system must provide a full transit itinerary consisting of either combination of public transport and walking.
- 3.3. The system must display detailed public transit information in each route result.

Train Information

- 3.4. The system must retrieve MRT information through external APIs to modify ETA accordingly.
- 3.5. The system must display the total number of train stops from start to end destination.
- 3.6. The system must display the duration of the train journey in hours and minutes.
- 3.7. The system must display train lines involved including necessary line transfers.
- 3.8. The system must provide detailed train platform specific information, including departure/arrival times, platform numbers.
- 3.9. The system must retrieve and display real-time platform density or crowd levels at relevant bus stops and train stations measured by “Low, Medium, High”.

- 3.10. The system must be able to retrieve MRT service disruption information and provide alternate public transport route suggestions.

Bus Information

- 3.11. The system must display bus service numbers in each route.
- 3.12. The system must display the total number of bus stops for each bus service.
- 3.13. The system must display the estimated length and duration of each bus journey.
- 3.14. The system must display estimated wait times at all bus stops within the route at their respective future timings.
- 3.15. The system must be able to retrieve traffic conditions and incidents and modify ETA accordingly.
- 3.16. The system must be able to suggest alternate public transport route suggestions when met with either traffic incidents or poor traffic conditions. (Traffic jams etc.)

4. Walking Mode

- 4.1. The system must provide an ETA based on the generated route.
- 4.2. The system shall display the estimated walking time in minutes.
- 4.3. The system must retrieve weather data from an external API.
- 4.4. The system must use weather information and display to the user current weather conditions.

5. User Interface and Experience

- 5.1. The user shall be displayed 4 route tabs in a structured format as:
 - 5.1.1. Best, Driving, Public Transit, Walking, where each tab will include step-by-step instructions and relevant information for each route. (e.g., carparks, transit stops, ETA).
 - 5.1.2. The number of routes for each tab are API dependent. The best tab shall display the shortest and most convenient route to the user.
- 5.2. The user shall be provided an interactive map interface to visualize routes for all transport modes.
- 5.3. The system shall display the best route above the tabs with the highest convenience score.
- 5.4. The Driving, Public transport, and walking tabs must display routes for each mode.

6. Best Result and Convenience Score

- 6.1. The system must calculate a convenience score for each route based on weighted factors dependent on the mode of transport. (Refer to Appendix A).
- 6.2. The system shall display the best route as a separate tab with the highest convenience score.

- 6.3. The system must allow users to prioritise factors to include within the convenience score for each mode of transport.
- 6.4. The system must generate a 'Best Result' tab displaying the top three routes in order (descending) and indicating their score.
- 6.5. The scoring system must be conveyed to the user through a clickable information icon which describes how selecting factors affect ranking priority.

7. User Login

- 7.1. The system must display a login form when a user clicks on a login button.
 - 7.1.1. The login form must contain the following fields:
 - Email
 - Password
 - Forget Password
 - Login Button
- 7.2. The system must locally validate user login input before submission.
 - 7.2.1. The system should check email validity. (must have an @ symbol).
- 7.3. The system must query the database attached to the system to verify credentials. At no point should there be any authentication with no connection to the account database.
- 7.4. The system must display an error message upon typing invalid credentials.
- 7.5. The system must implement a lockout system upon failure to provide valid credentials 5 times.
- 7.6. The system after 5 unsuccessful login attempts must send an email to the account's registered email of suspicious activity.
- 7.7. The system must prevent login and display authentication service unavailable when login services are down. (database uncontactable / other errors).
- 7.8. The system must provide error messages for invalid login attempts. (Wrong email / password).
- 7.9. The system must contain a clickable forgot password option button:
 - 7.9.1. The system must display a form to enter registered email.
 - 7.9.2. The system must be able to send a password reset link to the registered email.
 - 7.9.3. The system must be able to modify the account database and credentials upon receiving a successful and valid password reset attempt.

8. User Registration

- 8.1. The system must display a registration form when a user clicks on a login button.
- 8.2. The login form must contain the following fields:
 - First Name
 - Last Name
 - Email
 - Password

- 8.3. The system must validate user input before account creation submission.
- 8.4. The system must enforce that the password must contain at least 8 characters and at least one numerical AND one special character.
- 8.5. The system must enforce email to be in a valid format (presence of @ symbol).
- 8.6. The system must validate that password and password confirmation fields are the same.
- 8.7. The system must reject HTML tags, or any illegal characters deemed unsafe for security.
- 8.8. The system must display an inline error message under their respective fields when credentials do not meet the requirement.
- 8.9. The system must display a success message upon successfully creating an account and send an email to the registered email on registration status.

9. API Integration

- 9.1. The system must integrate with publicly available APIs for real-time data, such as:
 - LTA DataMall for public transit schedules, platform density, and traffic conditions.
 - data.gov.sg for carpark availability and rates. Mapping APIs such as OneMap for geocoding, routing, and map visualization.

10. Error Handling

- 10.1. The system must handle API errors gracefully and display a message in plain language (no technical jargon) such that more than 90% of users correctly interpret the meaning when data is unavailable.
- 10.2. The system must display an error message with an associated error code on an error page when faced with website related problems (page not found / error has occurred etc).
- 10.3. The system shall handle external API unavailability gracefully by showing a timeout message within 5 seconds, ensuring the UI is not frozen.

Non-Functional Requirements

1. Usability

- 1.1. The web application shall have a tutorial for first-time users on the functionality of the application
 - 1.1.1. The tutorial has an expected completion time taking no more than 5 minutes

- 1.2. The user interface must primarily be mobile-friendly, and can be accessed on both mobile and desktop browsers.
- 1.3. The navigation bar and other common elements between screens must remain consistent.

2. Performance

- 2.1. Routes must be displayed within 5 seconds of user input under normal network conditions.
- 2.2. The web application must be able to handle up to 100 concurrent users without significant performance degradation.

3. Security

- 3.1. HTTPS must be used to encrypt all data transmitted between the client and server in transit or in storage.
- 3.2. All user inputs must be sanitized to prevent potential cyber attacks (XSS, SQL injection, etc).
- 3.3. Codebase must not directly contain private API keys, passwords, or other confidential information.
- 3.4. The system should terminate user sessions when idle for more than 30 minutes in the application.

4. Compatibility

- 4.1. The system must be compatible with modern web browsers (e.g. Chrome, Firefox, Safari, Edge) on both desktop and mobile devices.

5. Logging and Maintainability

- 5.1. Logging and monitoring must be included to track errors and performance issues for debugging purposes.
- 5.2. System logs must be stored and viewable for at least 100 to 200 days since an incident.
- 5.3. Terms of use and query rate limits imposed by external APIs must be respected.

Data Dictionary

Term	Definition
User	The primary actor, typically a commuter in Singapore, who uses the SG Transport Planner Web App to plan routes in Singapore.
External APIs	Refers to APIs provided by external sources such as LTA Datamall, data.gov.sg and OneMap. These APIs provide real-time data for routes, traffic, carparks, and transit schedules required for the web application to function.
System	The backend of the web application
System Log	A recorded history of events within a computer or transport system.
Data Error (API Error)	An error code generated regarding external API access
Error Code	An identifier that indicates a specific problem in a system
Convenience Level Score	Refer to the Appendix A
ETA	Time taken to arrive at destination in minutes
User Profile	A personal profile to store information, preferences, and history including saved routes.
Interactive Map	An interactive map where users can pan and view points of interest.
Route	The directions from source to destination location based on the transport mode selected
Alternate Route	An alternate path to reach the same destination
Start, End Destination	Location of starting point and ending points taken from user's input
Travel Time	Time taken to travel from start to end destination.
Transport Mode	The type of transport to take. Available options include: 1. Driving 2. Public Transit 3. Walking

Peak Hour	The time when traffic is most crowded, more specifically 7AM - 9AM and 6PM - 8PM
Driving Mode	A transport mode where the person uses a private motor vehicle (e.g., car, motorcycle).
Carpark	A designated area where users can park their vehicles.
Traffic Incident Information	Information on traffic incidents that happen along the route which impact the route.
Traffic Disruption	An event where traffic is disrupted either by traffic incidents or by traffic congestion.
Public Transit Mode	Category of transportation service that is publicly available to multiple passengers, e.g bus and MRT.
Bus Wait Time	Waiting time at bus station
MRT Service Disruption	An event where Train Service along a certain line is disrupted, affecting ETA and route suggestion capability.
Station Crowded Level	A measure of how many people are present at a station within a given time (Low / Med / High)
Valid Location	Existing accessible local area in Singapore provided by external API
Carpark Availability	Presence of vacancies in each carpark lot.

Use Case Descriptions

Use Case ID :	1.1		
Use Case Name:	Search Location Auto-Suggest		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User, OneMap API
Description	Enables users to search for Singapore locations using a search bar with auto-suggest functionality. The system will query the OneMap Search API based on the user's input and display a dropdown list of possible matching locations to facilitate selection of start and end points
Preconditions	1. The system has access to the OneMap Search API 2. The user is able to access the search bar interface
Postconditions	The system returns a list of auto-suggested locations based on the user's input via a dropdown
Priority	High
Frequency of Use	High
Flow of Events	1. The user enters a partial or full location name in the search bar (e.g. "Orchard" or "Changi Airport") 2. The system sends the user's input to the OneMap Search API in real-time as the user types 3. The OneMap Search API returns a list of matching Singapore locations based on the input 4. The system displays the list of locations in an auto-suggest dropdown below the search bar
Alternative Flows	AF-S3: No Matching Locations 1. If the OneMap Search API returns no results for the user's input, the system will display an empty auto-suggest dropdown
Exceptions	API Error: 1. If the OneMap Search API is unreachable, the system will display an error message: "Unable to fetch location suggestions. Please try again later." in the auto-suggest dropdown
Includes	None
Special	None

Requirements	
Assumptions	None
Notes and Issues	None

Use Case ID :	1.2		
Use Case Name:	Validate Location		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User, OneMap API
Description	Validates that the user's submitted start and end locations are valid Singapore locations by querying the OneMap Search API to retrieve their coordinates. The locations are provided as text input in a search bar, and validation ensures they correspond to recognisable Singapore addresses
Preconditions	1. The system has access to the OneMap Search API 2. The user has entered start and end locations as text in the search bar
Postconditions	1. The system confirms the start and end locations are valid and retrieves their coordinates 2. The system continues to use case 1.3
Priority	High
Frequency of Use	High
Flow of Events	1. The user submits the start and end locations entered as text in the search bar 2. The system sends the submitted location text to the OneMap Search API 3. OneMap Search API returns the corresponding coordinates for the respective start and end location if valid 4. The system stores the coordinates and accepts the locations

	as valid 5. The system proceeds to use case 1.3
Alternative Flows	AF-S3: Incomplete or Ambiguous Location 1. If the submitted location text is incomplete or ambiguous (e.g. "Orchard"), the OneMap Search API will return multiple locations and their coordinates instead of a single location. In this case, the system will choose the first location amongst all locations returned by the OneMap Search API
Exceptions	Invalid Location: 1. If the OneMap Search API returns no locations based on either the user's start and end location input, the system displays an error message: "Invalid location. Please enter a valid Singapore location and try again." API Error: 1. If the OneMap Search API is unreachable, the system will display an error message: "Unable to validate locations. Please try again later."
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.3		
Use Case Name:	Select Mode of Transport		
Created by:	Andy Thiang	Last updated by:	Andy Thiang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	Allow the user to select between modes of transport
Preconditions	Start and end location validated

Postconditions	Display the information related to the mode of transport selected by the user
Priority	High
Frequency of Use	High
Flow of Events	1. User enters the main page which displays the multiple transport tabs (Best, Driving, Public Transport, Walking etc) 2. User clicks on the respective tab (mode of transport) 3. The system will query the external API (LTA data mall / onegov) for conditions as specified by the <u>convenience score</u> for all modes of transport regardless of the selected tab 4. The system displays the results with respect to the selected mode of transport. 5. Users can swap to other transport tabs and view 3 - 4 again.
Alternative Flows	AF-S3: 1. Data Unavailable (from external API) - External API does not return any routes for a specific transport mode (network or invalid) 2. Tab will display error message "No routes available"
Exceptions	1. The system cannot connect to the external API server. 2. The system will display an error message, "Connection Error."
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.4		
Use Case Name:	Select Convenience Filter		
Created by:	Benjamin Phang	Last updated by:	Benjamin Phang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	The system will display a drop down of filters which the user can select. These filters modify and directly affect convenience score in turn affecting routes displayed to the user.

Preconditions	Start and end location validated
Postconditions	1. The system saves the convenience score filter preference
Priority	High
Frequency of Use	High
Flow of Events	1. The system will display a drop down of filters selectable by the user (all filters are selected by default) 2. The user can select / deselect singular, multiple, or no convenience filters 3. The system will display a checkmark on user selected filters 4. The system stores the user's convenience score filter preference for further calculations
Alternative Flows	None
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	1.5		
Use Case Name:	Display Convenience Level Score		
Created by:	Benjamin Phang	Last updated by:	Benjamin Phang
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
Description	The system will calculate and display a convenience level score for each mode of transport. The score will be within the range of 1-10, and they will be displayed to the user by rank based on the convenience level score
Preconditions	Start and end location validated
Postconditions	1. Convenience score displayed per route 2. "Best" tab indicating the top three routes by convenience score
Priority	High

Frequency of Use	High
Flow of Events	1. The system will retrieve the user's selected convenience filters 2. Convenience score is calculated for all routes within the range of 1 - 10 3. Convenience score is displayed to the user on each route card 4. The top three convenience scores from all transport modes are ranked in descending order in a separate "Best" tab
Alternative Flows	None
Exceptions	None
Includes	1.4
Special Requirements	None
Assumptions	External API provides accurate and updated route data
Notes and Issues	None

Use Case ID :	1.6		
Use Case Name:	Display Interactive Map		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, One Map API
Description	User to be provided an interactive map interface which allows users to interact with such as being able to click on locations
Preconditions	Users must have an internet connection to be able to connect to the application and API is working
Postconditions	The interactive map interface has been generated and the user is able to see the map of Singapore
Priority	High
Frequency of Use	High
Flow of Events	1. The user opens up the app. 2. Upon opening the application, the main UI will request for OneMapAPI data. 3. API will receive the request and return the main UI with map

	data. 4. The Main UI will receive the data and display the full map of Singapore with the ability to interact with it.
Alternative Flows	AF-S2: Interactive map is not displayed 1. The user opens the application but the interactive map does not load in, the user will reopen the application.
Exceptions	API Error 1. After Starting Point and Destination addresses are inputted 2. The Application goes to External API to retrieve info but is met with an error with the API 3. Displays an Error message to inform the user that the API is currently experiencing troubles (API system not functional, API unable to determine information given, etc.)
Includes	None
Special Requirements	None
Assumptions	Assumes that Application will only be used in Singapore
Notes and Issues	Locations are that not in the system of the external API will not have information on the interactive map

Use Case ID :	1.7		
Use Case Name:	Retrieve Routes		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, OneMap API
Description	This use case retrieves route options from the OneMap API based on the user's start and end location
Preconditions	1. The system has access to the OneMap API 2. Start and end location have been validated
Postconditions	The system has retrieved and prepared route options for the selected mode of transport.
Priority	High
Frequency of Use	High
Flow of Events	1. The system sends a request to the OneMap API with the

	coordinates of the start and end location 2. The OneMap API responds with route options 3. The system processes the response and prepares the route data for display. 4. The system continues to either use case 2.1, 3.1, 4.1
Alternative Flows	None
Exceptions	API Error: 1. If the OneMap API is unreachable, the system will display an error message: "Unable to retrieve routes. Please try again later."
Includes	None
Special Requirements	None
Assumptions	Assumes User has selected Starting Point and Destination
Notes and Issues	None

Use Case ID :	1.8		
Use Case Name:	Save Routes (Itinerary)		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User
Description	This interface will show the user their saved/favourited routes for ease of use if this is a common route they take
Preconditions	User is logged into their account and has chosen a Starting Point and Destination with the routes and itinerary generated
Postconditions	User has picked a route and saved it to Saved Routes interface
Priority	Low
Frequency of Use	Low

Flow of Events	1. After user has been displayed generated itinerary, user will click the 'Save' button next to the route in the route card 2. Itinerary will be saved to the Saved Routes interface for future reference or use 3. User will be able to access their saved route through the saved routes tab 4. User can rename the routes (ie. Home to School)
Alternative Flows	AF-S3 No Route option saved 1. If the user did not save any routes previously, system will display "No Saved Routes" 2. User to select a Starting Point and Destination 3. Users can choose their desired route and click the bookmark (save) icon. AF-S3 Duplicate Route 1. If the user upon clicking the saved route already has a route with the exact same path and details, display an error message "route already exists!" AF-S3 User not Logged in 1. If the user is not logged in, the system will display the login pop up prompt. 2. Upon successful login, the user may press on the save route button again
Exceptions	None
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	The route saved will only take the time taken and convenience level of the time before the route was saved. Saved routes will update present time when the saved tab is clicked

Use Case ID :	1.9		
Use Case Name:	Interactive Map Start/End Point Selector		
Created by:	Hong Ming	Last Updated By:	Hong Ming
Date	1/9/2025	Date Last Updated:	1/9/2025

Actor:	User
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Description	User can pick their Starting Point or Destination by clicking directly on the interactive map
Preconditions	The map is displayed and clearly visible to the user
Postconditions	System receives start and end location data to be validated.
Priority	Low
Frequency of Use	Low
Flow of Events	1. User will click on a location on the interactive map 2. Information that is related to the location will pop up as a bubble (Postal Code, Street Name, Building Name) 3. User selects whether this location type (start / end) 4. Application will automatically fill in the respective position
Alternative Flows	AF-S1 Location is invalid point in the map 1. User clicks on a location which has no valid coordinate 2. Display Error Message
Exceptions	Interactive map API is not unreachable
Includes	1.1, 1.2, 1.6
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID:	2.1		
Use Case Name:	Display Driving Route		
Created By:	Dinh	Last Updated By:	25/08/2025
Date:	25/08/2025	Date Last Updated:	25/08/2025

Actor:	User, One Map API
Description	Display route results in driving mode.
Preconditions	User input start and end destination and select driving as preferred mode of transport.
Postconditions	The system shall display route results as a “route card”

Priority	Medium
Frequency of Use	High
Flow of Events	1. The user selects driving as the preferred mode of transport. 2. The system retrieves the list of available driving routes from the parent use case Retrieve Routes. 3. The system displays route cards for each driving route (e.g., including distance, travel time). 4. The system plots the driving routes on the interactive map. 5. The user may select a route card to view detailed instructions
Alternative Flows	AF-S3: 1. If there are no possible driving routes, the system shall inform the user and propose to switch to other modes of transport.
Exceptions	None
Includes	1.5, 2.2, 2.3
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID:	2.2		
Use Case Name:	Display Nearby Carparks Info		
Created By:	Dinh	Last Updated By:	25/08/2025
Date:	25/08/2025	Date Last Updated:	25/08/2025

Actor:	User, Government Data API
Description	Display nearby carparks info in driving mode.
Preconditions	User input start and end destination and select driving as preferred mode of transport.
Postconditions	Information about carparks near to the end destination is displayed.
Priority	Medium
Frequency of Use	High
Flow of Events	1. The user selects driving as mode of transport.

	- The system gets driving routes by querying the external API. - The system retrieves possible routes from external API. - For each route, the systems retrieve information about nearby car parks such as availability, parking rate, etc. - The system displays nearby car parks as pins in the map . - If users click on the pins, a window pops up showing the detailed information about the carpark.
Alternative Flows	AF-S6: If there is a lack of data about some data like parking rates, the system displays that the information is not available.
Exceptions	EX1: If the API is down, The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	2.1
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	2.3		
Use Case Name:	Display Traffic Incident Info		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, LTA DataMall API
Description	Retrieve traffic incidents information on the route.
Preconditions	User input start and end destination.
Postconditions	Information about traffic incidents on the route is displayed.
Priority	Medium
Frequency of Use	High
Flow of Events	The system gets possible routes by querying the external API.

	- For each route, the system queries external API and filters incidents that directly impact the route. For public transport mode, the system queries MRT breakdowns. For driving mode, the system queries traffic accidents on the road. - The system displays that information on the map with pins on the affected route segments and names of roads affected on the route result card.
Alternative Flows	AF-S3: No incident found, the system does not display any pins on the route.
Exceptions	EX1: If the API is down, The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	2.1, 2.2
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	3.1		
Use Case Name:	Display Public Transit Route		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, One Map API
Description	Show buses/trains needed with transfer points.
Preconditions	User input start, end destination and select public transport mode.
Postconditions	Information about traffic incidents on the route is displayed.
Priority	High
Frequency of Use	High
Flow of Events	The user selects public transit as the preferred mode of transport.

	- The system retrieves the list of available public transit routes from the parent use case Retrieve Routes. - The system displays route cards for each public transit route (e.g., including distance, travel time). - The system plots the public transit routes on the interactive map. - The user may select a route card to view detailed instructions such as buses and MRT to take as well as walking segments
Alternative Flows	AF-S2: - Services are unavailable due to the time of day, then the system displays that error to the user. AF-S1: - User input location outside of Singapore, the system then displays the message to the user.
Exceptions	None
Includes	1.5, 2.3, 3.2
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	3.2		
Use Case Name:	Display Public Transit Info		
Created by:	Dinh	Last Updated By:	Dinh
Date	25/8/2025	Date Last Updated:	25/8/2025

Actor:	User, LTA DataMall API
Description	Display detailed information about platforms and crowd level.
Preconditions	User input start, end destination and select public transport mode.
Postconditions	Information is displayed to the user.
Priority	High

Frequency of Use	High
Flow of Events	1. User select a particular itinerary in the public transit mode. 2. For each route in the itinerary: - System retrieves scheduled departure/arrival. - System retrieves real-time updates such as delay or breakdown, platform density, estimated travel time. - Platform density is displayed as “Low, Med, High” 3. API returned itineraries containing buses, MRT and walking segments.
Alternative Flows	AF-S2: 1. Platform information is missing, the system show the information is not available.
Exceptions	EX1: If the API is down, 1. The system informs the user that the service is temporarily unavailable and includes a Retry action.
Includes	None
Special Requirements	None
Assumptions	None
Notes and Issues	None

Use Case ID :	4.1		
Use Case Name:	Display Walking Route		
Created by:	Andy	Last Updated By:	Andy
Date	26/8/2025	Date Last Updated:	26/8/2025

Actor:	User, OneMap API, Government Data API
Description	Shows Walking route
Preconditions	User input start and end destination and select walking as preferred mode of transport.

Postconditions	The system shall display walking route results as a “route card”
Priority	High
Frequency of Use	High
Flow of Events	1. The user selects public walking as the preferred mode of transport. 2. The system retrieves the list of available walking routes from the parent use case Retrieve Routes, and the weather forecast. 3. The system displays route cards for each walking route (e.g., including distance, travel time, weather forecast...). 4. The system plots the walking routes on the interactive map. 5. The user may select a route card to view detailed instructions on the walking directions
Alternative Flows	None
Exceptions	None
Includes	1.5
Special Requirements	None
Assumptions	Start and End location is validated
Notes and Issues	None

Use Case ID :	5.1		
Use Case Name:	User Login		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
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Description	The back and front end system involved in managing authentication services within the application. It is used in validating entered credentials and providing appropriate error messages otherwise
Preconditions	1. User must already have an existing account within application database 2. The system is already connected to the account database
Postconditions	1. User is assigned a login session token by the authentication service 2. The user is displayed an error message upon input of invalid credentials not present in the application's database
Priority	High
Frequency of Use	High
Flow of Events	1. The user clicks on the login button within the landing page 2. The system pops up a login form with email and password and a login and a forget password button 3. The system validates the input credentials (e.g valid email, password length) 4. The system queries and validates with the account database to check account validity 5. If account is valid, the system provides a login session token and redirects the user to the main page
Alternative Flows	None
Exceptions	1. Invalid Login Credentials - If the user provides invalid login credentials, an error message with the text "invalid login credentials, please try again" message below the associated fields 2. Attempt Lockout - After 5 failed login attempts, the account is temporarily locked and an email is sent to the user's email on suspicious login activity 3. Authentication Service is Down - Due to maintenance or other related issues pertaining to the attached credential database, login services will be disabled and will display an error message. "Login services are unavailable at the moment, please try again later".
Includes	5.2
Special Requirements	1. The application in production must support HTTPS and have valid authority certificates. 2. The password in transit and in storage must be salted and hashed for data security
Assumptions	1. The user has an existing valid email
Notes and Issues	Optionally implement Google / external OAuth for login instead of creating an account with our system

Use Case ID :	5.2		
Use Case Name:	Register User		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	(Accountless) User
Description	Registration of a user with no existing account, providing credentials such as email address, password and username for added features of the app.
Preconditions	1. User is not logged in 2. User has an existing reachable email address
Postconditions	1. Account creation successful, updated in the system's credential database 2. Data stored is validated and sanitized to prevent against cross site scripting (XSS) 3. Confirmation email sent to email address
Priority	High
Frequency of Use	High
Flow of Events	1. User clicks on sign up button on the login screen 2. System will pop up a registration form with the following details - First Name - Last Name - Email Address - Password 3. System validates entered data with the following criteria a. Password i. 8 Characters ii. At least 1 numerical digit iii. At least 1 special symbol / character b. Email Address i. Presence of @ symbol 4. System sanitises and prevents encoding of HTML tags 5. Upon successful validation, create a new user and store it within the system's account database 6. The system displays "Account Creation Successful" and redirects the user to the main page.
Alternative Flows	AF-S3 Weak / Invalid Password / email - User inputs invalid password / email - System displays an error message below the respective field

	AF-S3 Account Exists - User inputs an email that already exists in the database - Upon pressing login, the system displays an error and suggest resetting password or to use another email
Exceptions	Invalid Characters - Reject html encoded tags such as <script> and provide an error message for illegal use of characters Authentication Services Down - Provide an error code and display the message "Authentication Services are unreachable, please try again later" message
Includes	None
Special Requirements	1. The application in production must support HTTPS and have valid authority certificates. 2. The password in transit and in storage must be salted and hashed for data security 3. The system must sanitise and encode all user inputs to prevent XSS. 4. Error messages must not reveal anything about the system
Assumptions	1. The user has an existing valid email
Notes and Issues	Consider using third party oauth like Google instead of directly registering an account.

Use Case ID :	5.3		
Use Case Name:	Reset Password		
Created by:	Caasi John	Last updated by:	Caasi John
Date	25th August 2025	Date Last updated:	25th August 2025

Actor:	User
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Description	Feature to allow for the resetting of passwords through an email link upon the user forgetting their account's password
Preconditions	1. Associated account to be reset within the application database must exist 2. The system is already connected to the account database
Postconditions	3. The user's password is updated in the database
Priority	High
Frequency of Use	High
Flow of Events	Forget Password Option 1. The user presses the forget password option on the pop up login form 2. The system will display a form which prompts users to input their email address 3. The system will send a password change request email to the user's input email address 4. The user enters a new password in a web form from the link given to their email 5. New password must meet password requirements and must match, otherwise prevent the register button from working 6. The system updates the account database on user credential change on success
Alternative Flows	None
Exceptions	1. Authentication Services Down - Provide an error code and display the message "Authentication Services are unreachable, please try again later" message 2. Invalid Characters / Not meeting minimum requirements - Reject html encoded tags such as <script> and provide an error message for illegal use of characters
Includes	5.1
Special Requirements	1. The domain of the sending email (email service that sends the reset link) must be of a unique and trusted domain
Assumptions	1. The user has an existing valid email in the database
Notes and Issues	None

Appendix A

Convenience Score Considerations

Related to All / Walking

- Time taken to reach the destination (More time, less convenient)
- Amount of Walking (More walking, less convenient)

Public Transport Related

- Number of transfers (More transfers, less convenient)
- Crowd Level at train station. (Denser population, less convenient)
- Bus wait time (Longer wait times, less convenient)

Driving Related

- Carpark Availability near destination (Less availability, less convenient)

Normalized time equation:

Each normalised value will be scaled by 10 (multiply by 10)

- Time taken to reach the destination (More time, less convenient)

$$\text{Normalized Time} = \frac{\text{Time Taken} - \text{Min Time}}{\text{Max Time} - \text{Min Time}}$$

- Amount of Walking (More walking, less convenient)

$$\text{Normalized Walking} = \frac{\text{Distance} - \text{Min Distance}}{\text{Max Distance} - \text{Min Distance}}$$

- Number of transfers (More transfers, less convenient)

$$\text{Normalized Transfers} = \frac{\text{Transfers}}{\text{Max Transfers}}$$

- Crowd Level at train station. (Denser population, less convenient)

$$\text{Normalized Crowd} = \frac{\text{Crowd Percentage}}{100}$$

- Bus wait time (Longer wait times, less convenient)

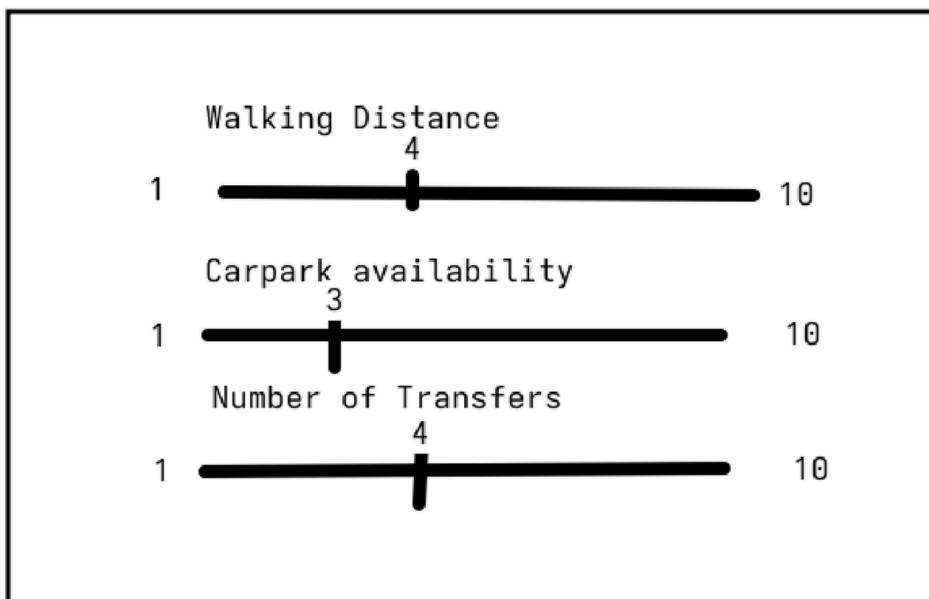
$$\text{Normalized Wait Time} = \frac{\text{Wait Time}}{\text{Max Wait Time}}$$

- Carpark Availability near destination (Less availability, less convenient)

$$\text{Normalized Carpark} = 1 - \frac{\text{Availability Percentage}}{100}$$

Calculate weightage:

The importance of each factor is determined between a value of 1 to 10 and is selectable by the user. For example, if the user selects the value 4 for walking distance, the value 3 for carpark availability and the value 4 for number of transfers, the weight of each factor is calculated using the formula shown below.



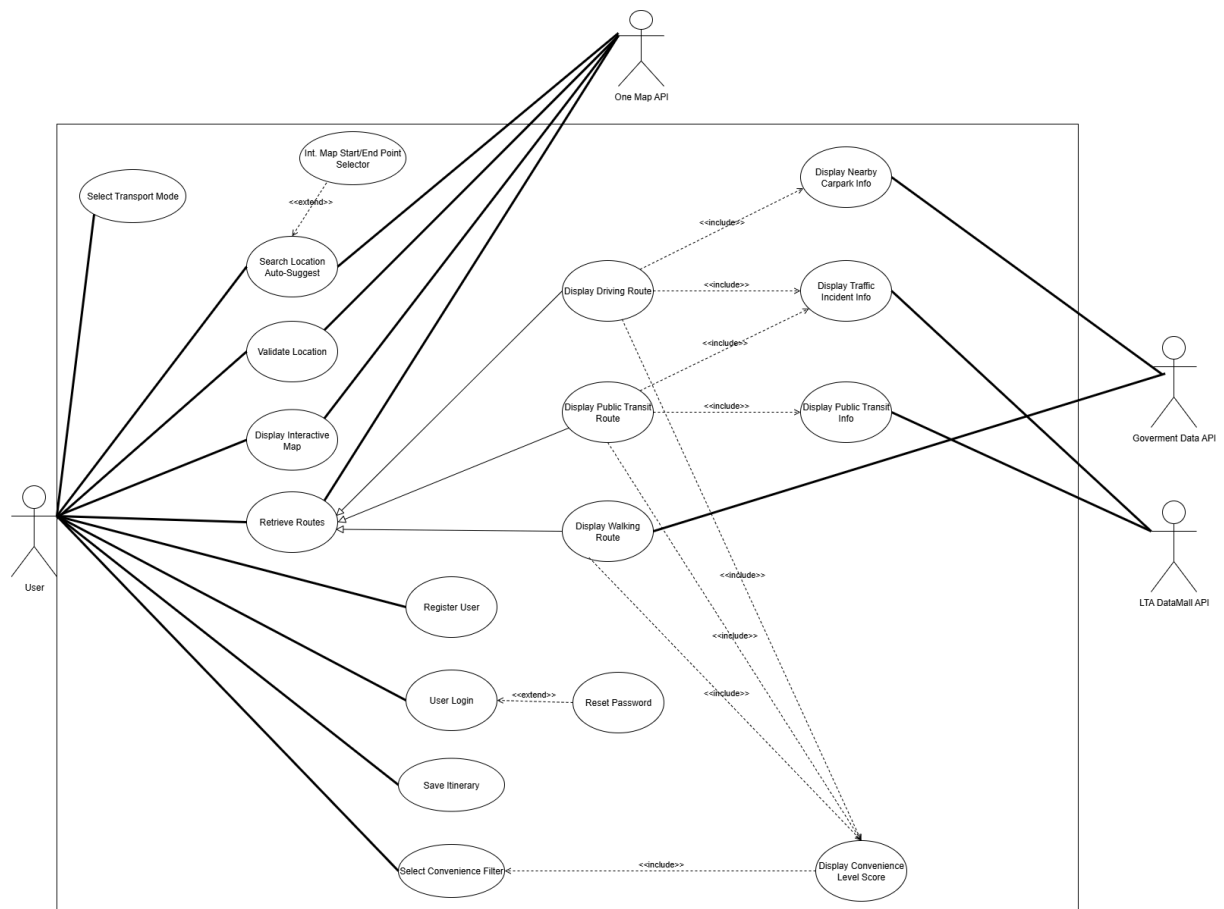
$$\text{Weight}(W_i) = \frac{\text{Selector Value}(S_i)}{\text{Sum of All Selector Values}(\sum S_j)}$$

Each normalised score will be multiplied by its respective weightage.

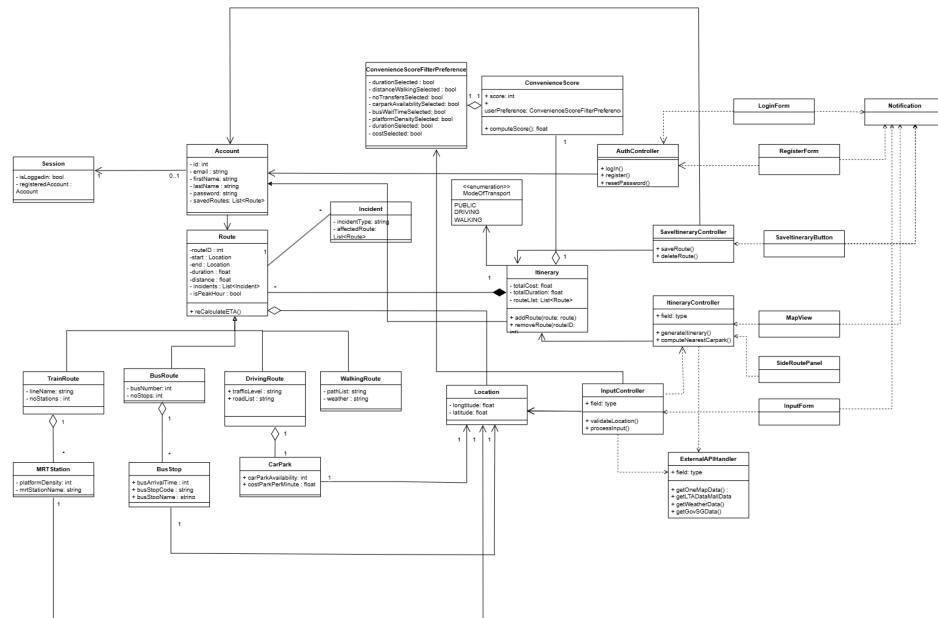
Convenience score:

All normalised scores will be added together to find the inconvenience score.
The convenience score will be (10 - inconvenience score).

Use Case Diagram

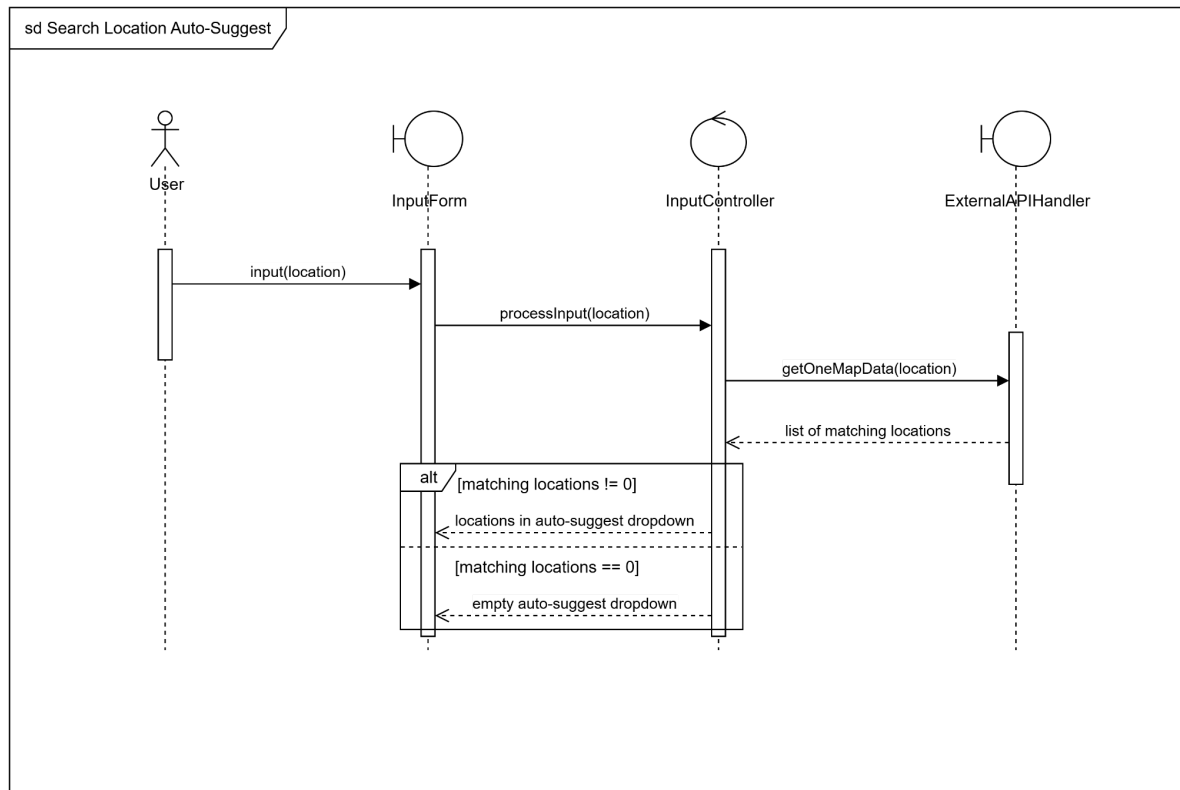


Class Diagram

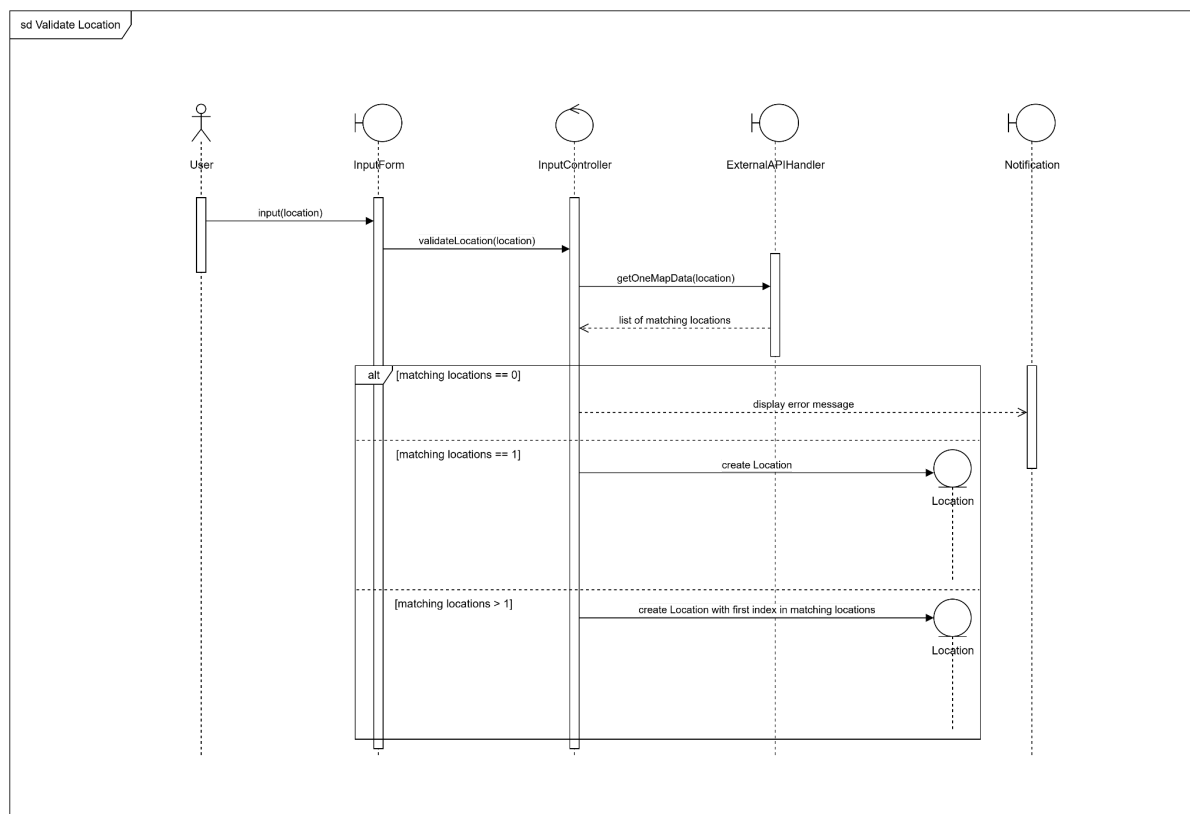


Stereotype Class Diagram

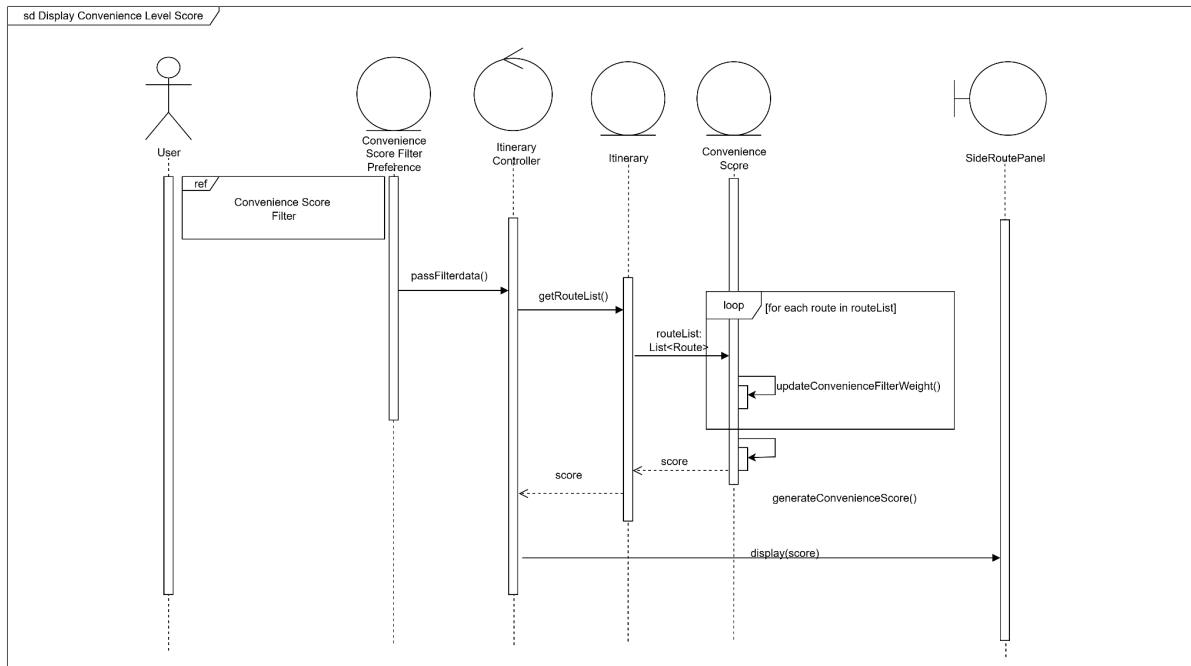
1.1 Search Location Auto-Suggest



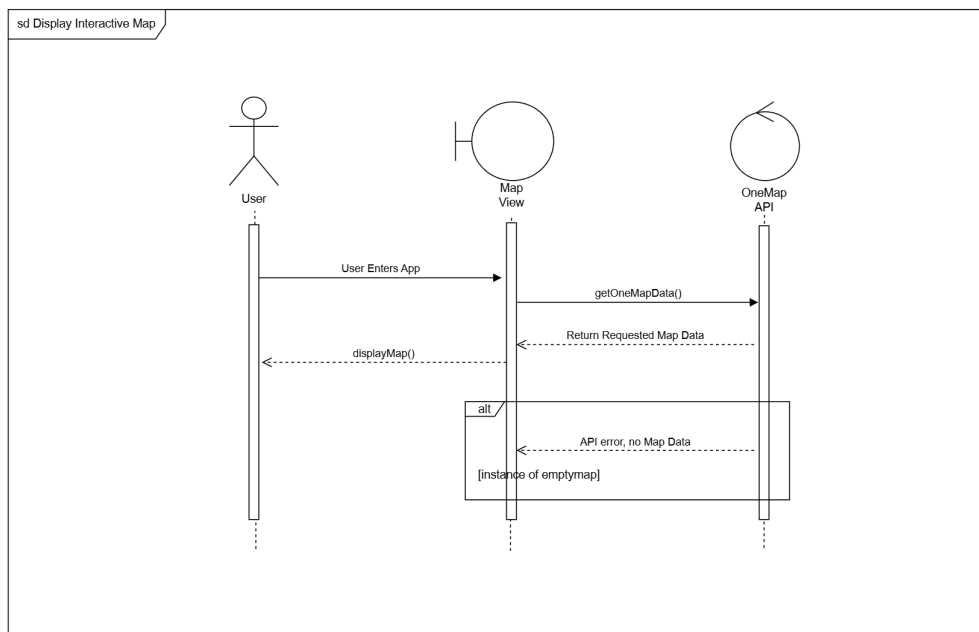
1.2 Validate Location



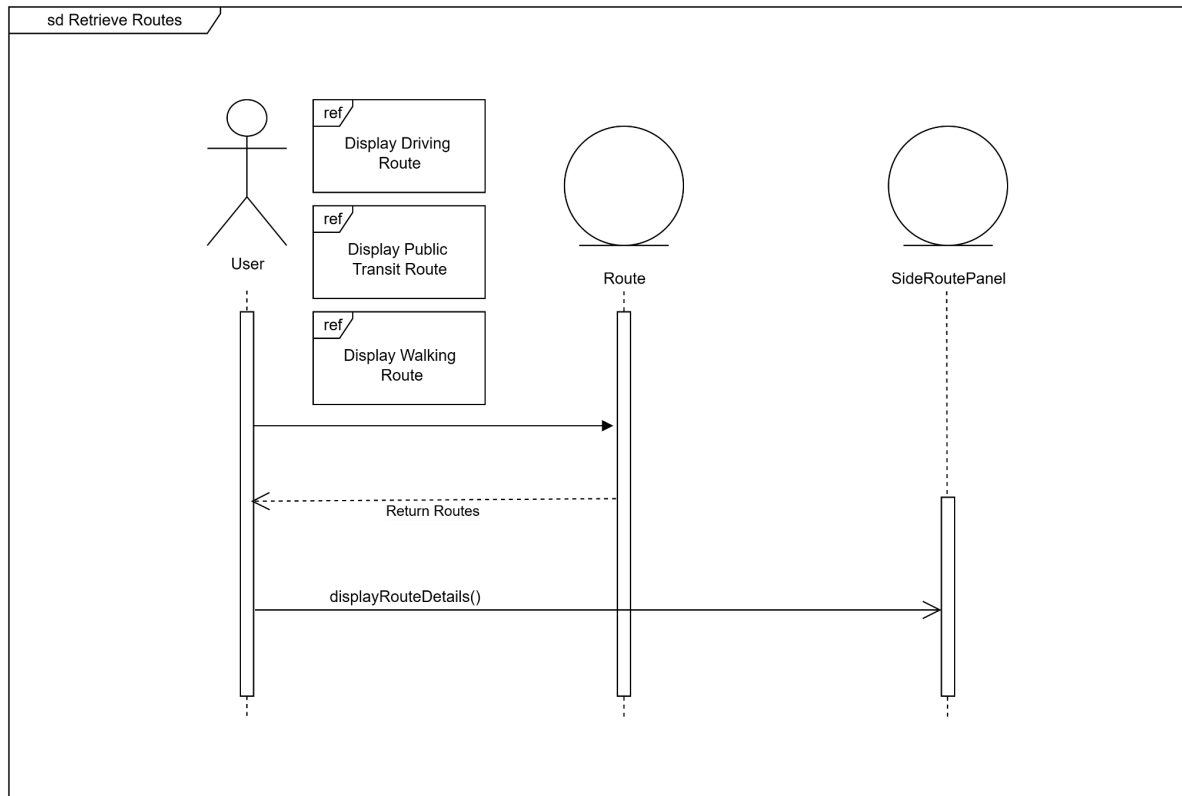
1.5 Display Convenience Level Score



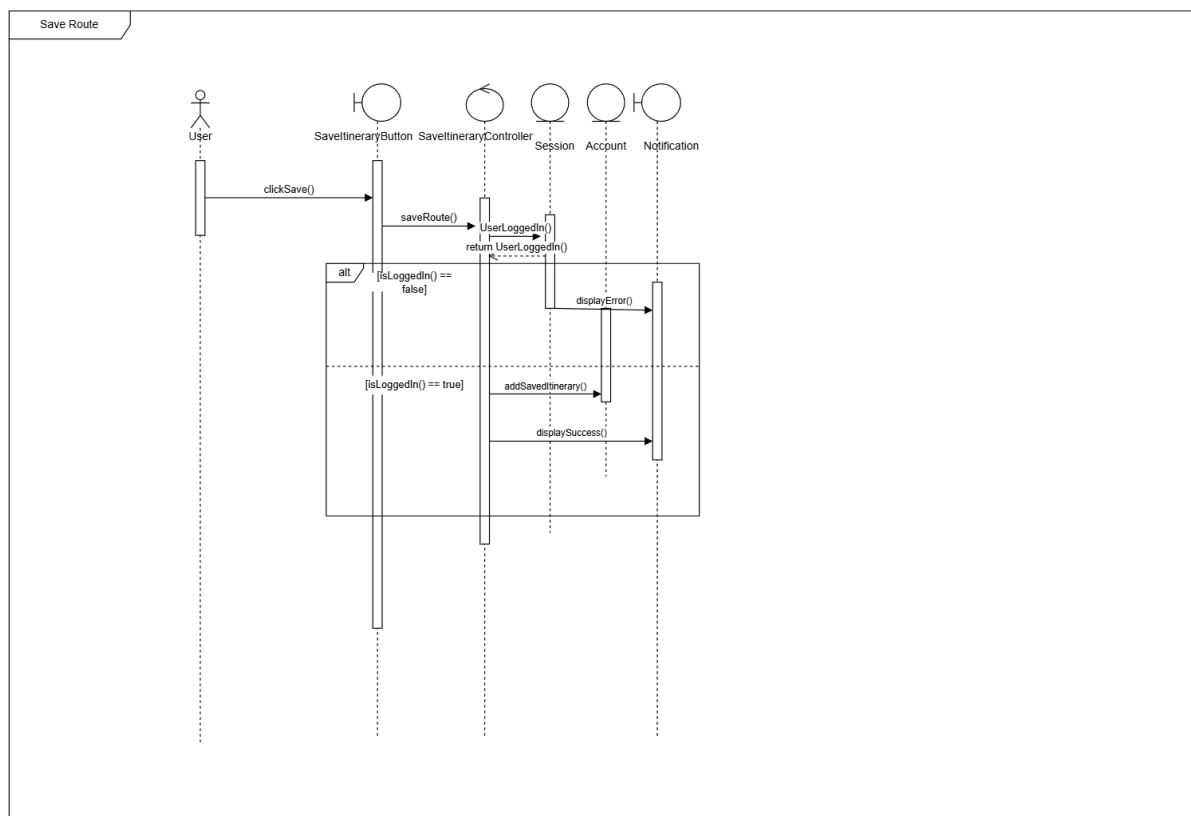
1.6 Display Interactive Map



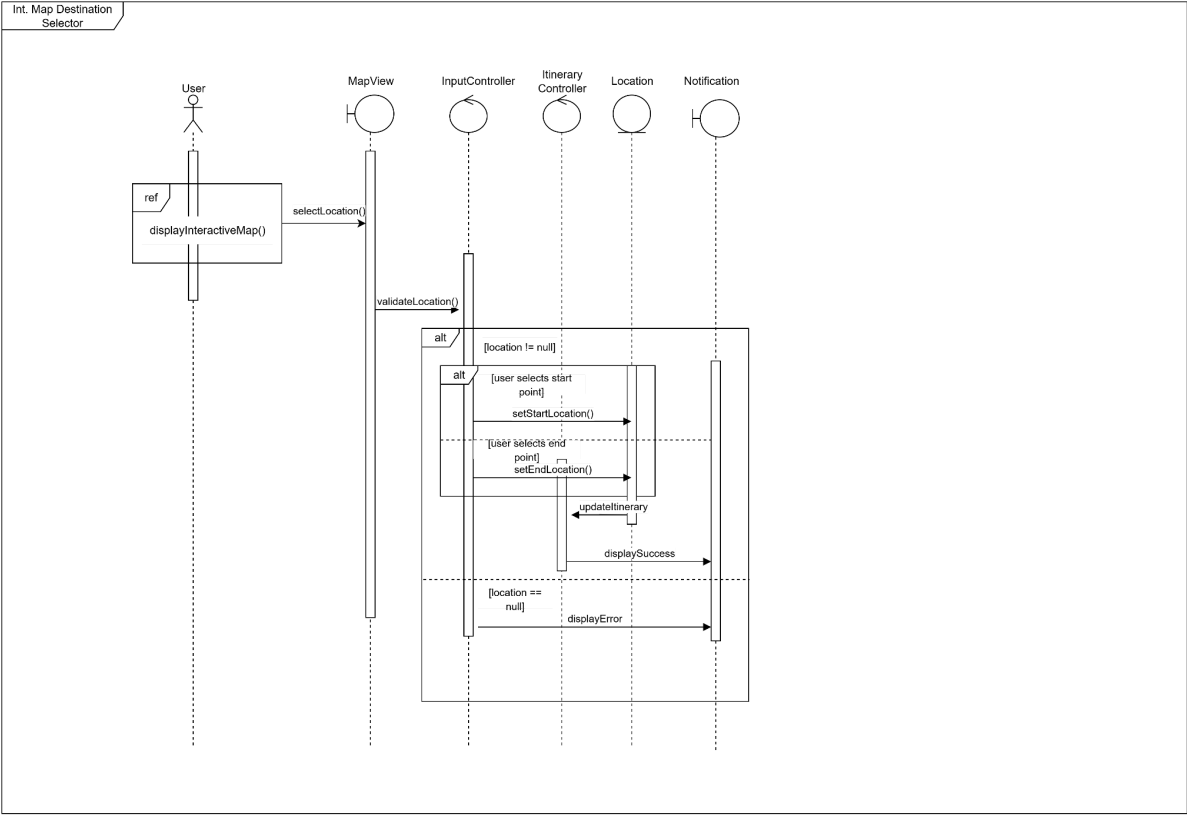
1.7 Retrieve Routes



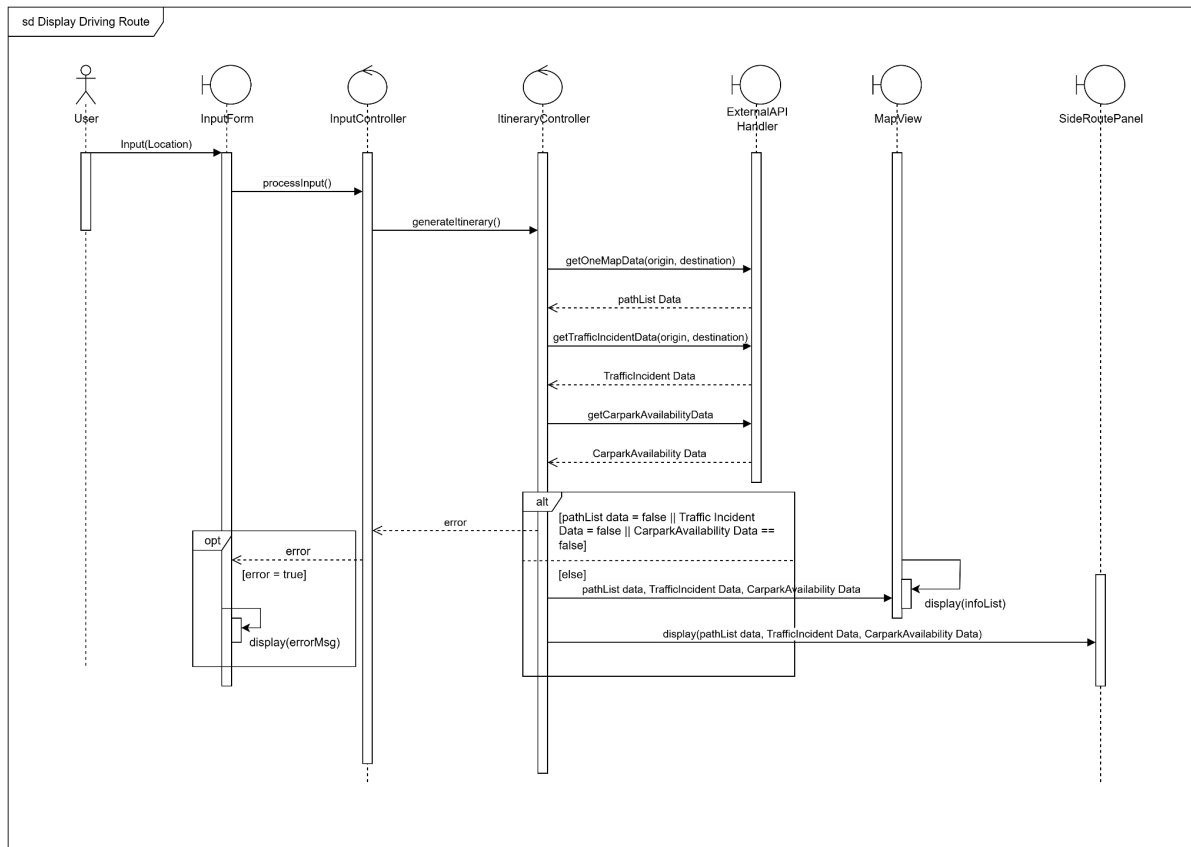
1.8 Saved Routes (Itinerary)



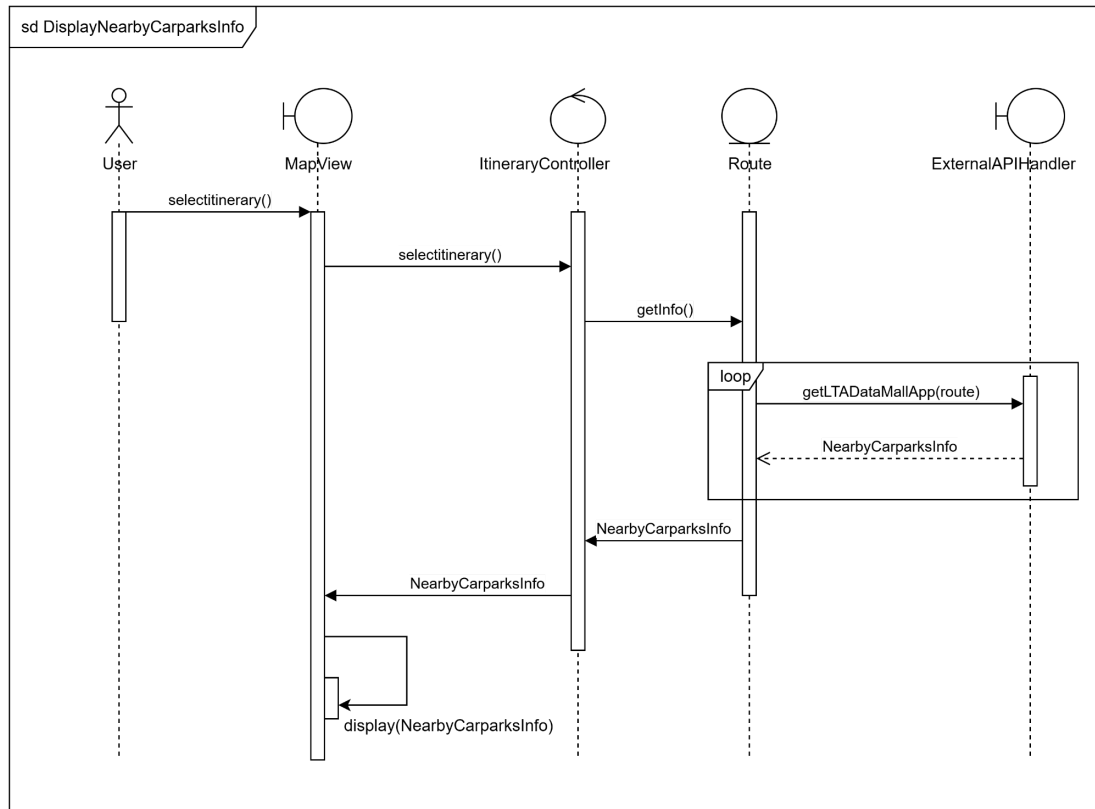
1.9 Interactive Map Destination/Starting Point Selector



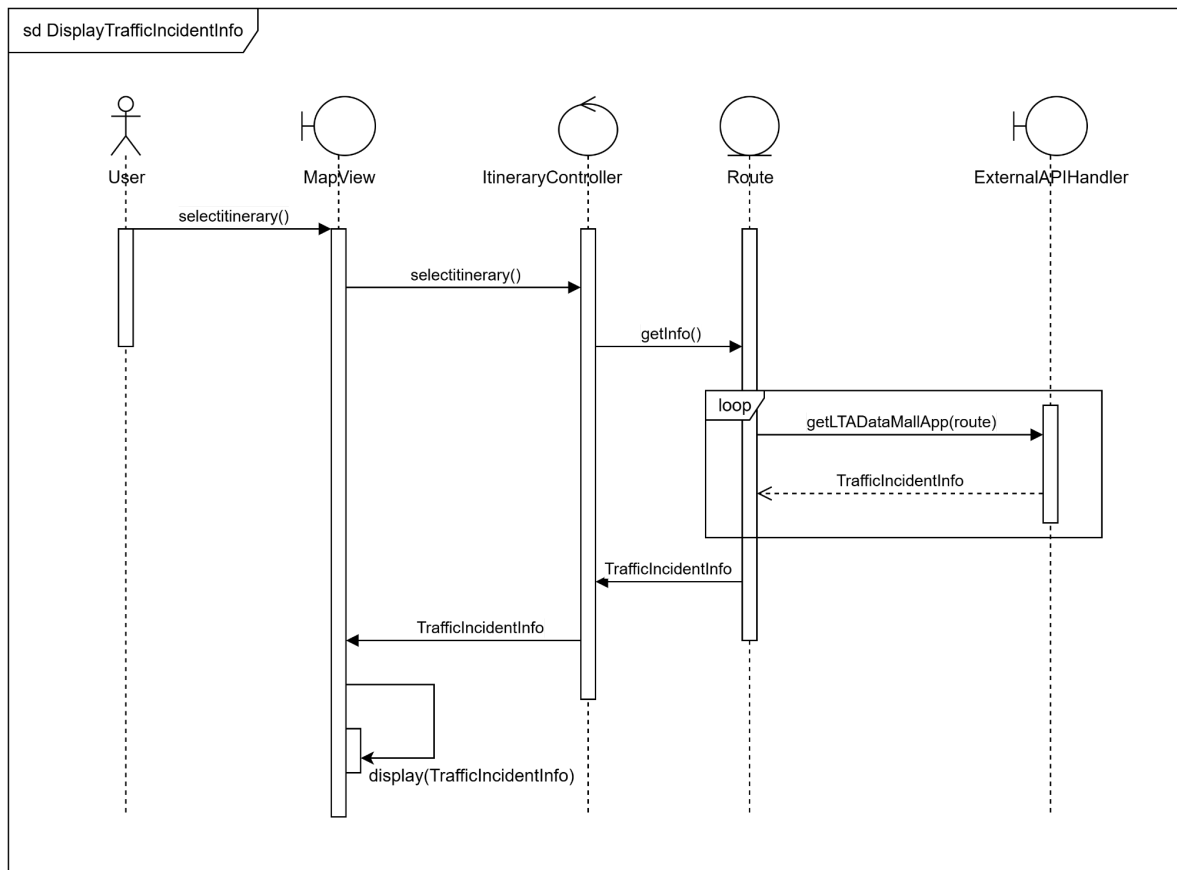
2.1 Display Driving Route



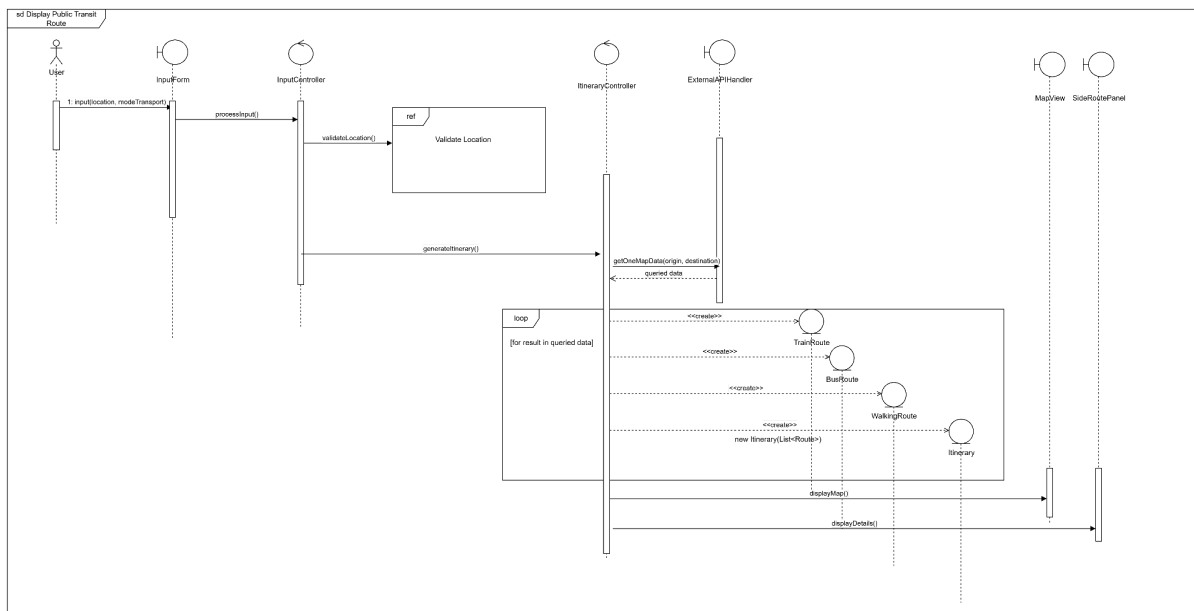
2.2 Display Nearby Carpark Info



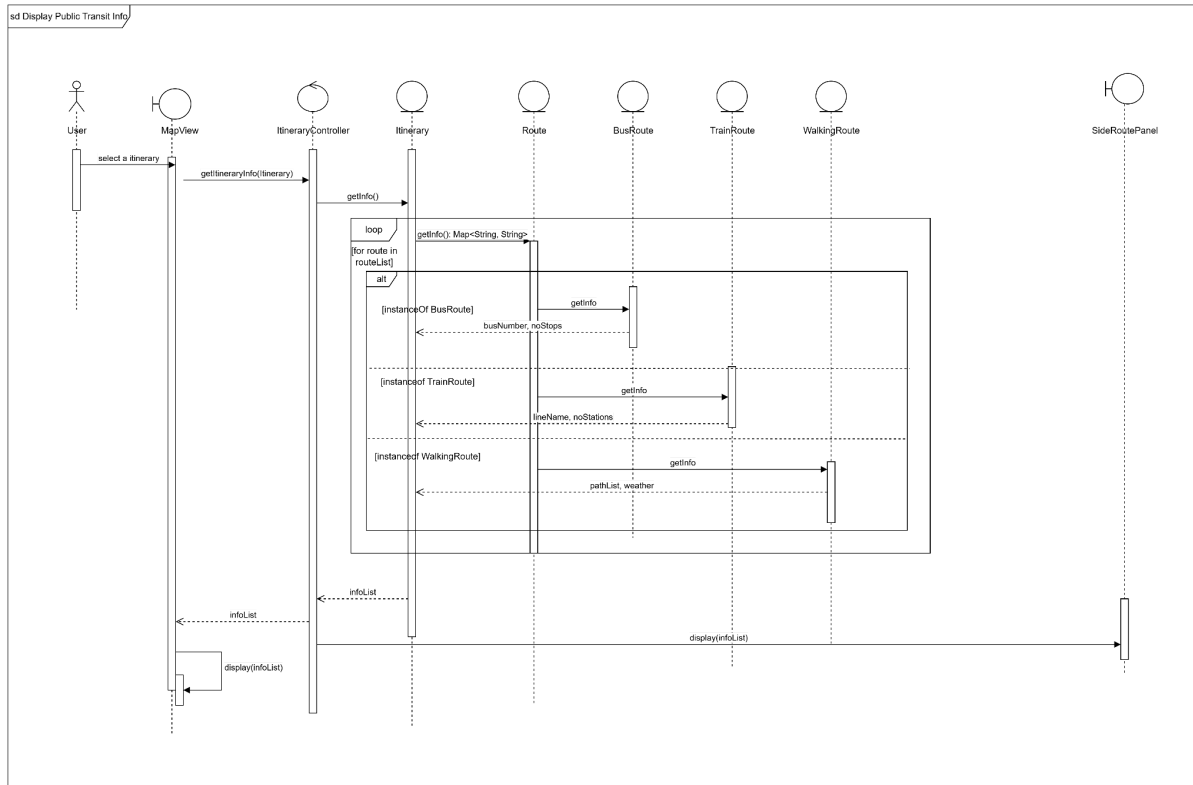
2.3 Display TrafficIncident Info



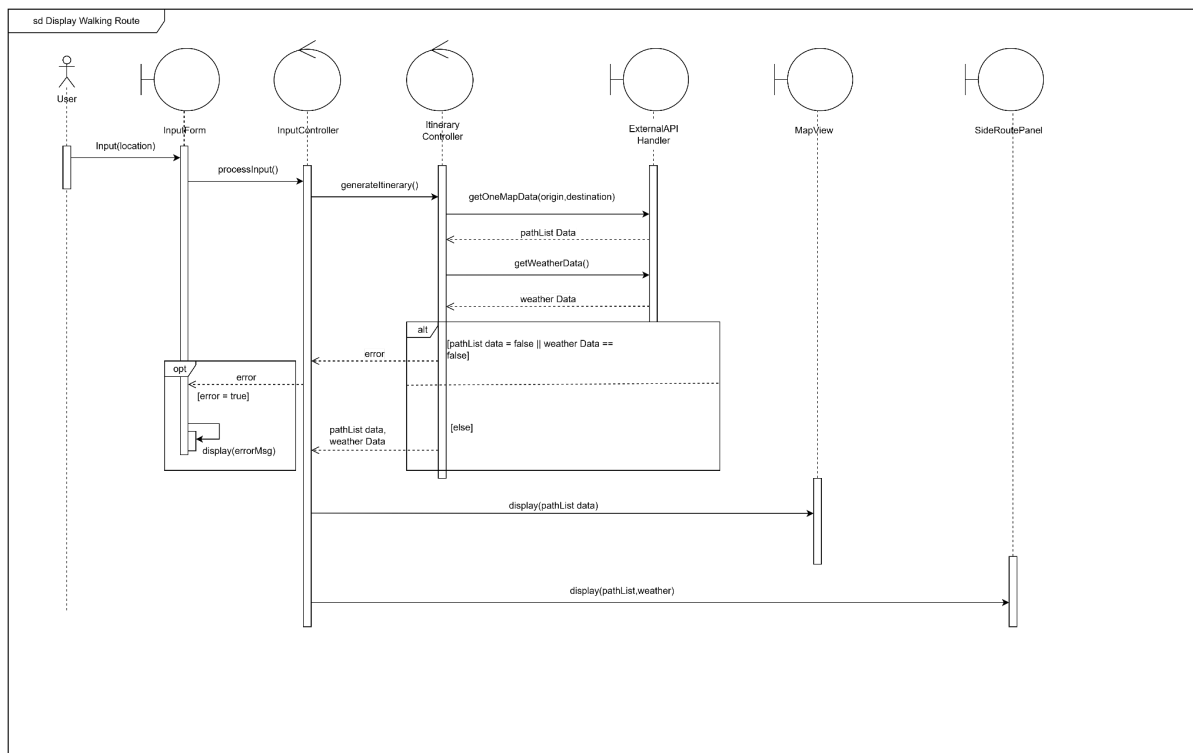
3.1 Display Public Transit Route



3.2 Display Public Transit Info



4.1 Display Walking Route



Dialog Map

